Mess Manager — PRD v1.2

Oct 13, 2026 · @Abu Uyoon

A Global, Transparent Shared-Expense Management Platform. Developed by Uyoon Tech Solutions (A Unit of Unifair Trading Pvt. Ltd., India). Founder: Imran Khan (Abu Uyoon).

0. Document Control

v1.2 locks the founder's decisions of 13 October 2026: in-app account deletion with a 2-month audit hold, mandatory payment proof with Admin confirmation, month lock after settlement, a dispute and flag system, Google Sign-In, an ad-supported free model, and a strictly neutral (no flags) interface.

Field	Value
Document Title	Mess Manager — Product Requirements Document
Version	1.2
Prepared For	Uyoon Tech Solutions (A Unit of Unifair Trading Pvt. Ltd., India)
Document Owner	Imran Khan (Abu Uyoon) — Founder
Status	Draft — pending founder sign-off on Section 28 (Open Decisions)

Revision History

Version	Date	Author	Change Summary
1.2	13 Oct 2026	Product Team	Account deletion lifecycle; Google Sign-In with privacy rules and App Lock; currency and time-zone change rules; split methods; payment proof and Admin confirmation; settlement month lock; dispute and flag system; offline sync safety rules; neutral localization; ads policy; legal and compliance section; MoSCoW corrected to match launch scope; Play Store publishing requirements
1.1	25 Jul 2026	Product Team	Added Executive Summary, Goals/KPIs, Scope, Personas, Prioritization, NFRs, Risks and Document Control
1.0	—	Uyoon Tech Solutions	Initial functional requirements draft

1. Executive Summary

Mess Manager is a free, ad-supported app that makes every shared expense in a group visible, provable and fairly split. It serves apartments, hostels, messes, families, offices and friend groups. Every financial action inside a group (a "Room") is recorded, traceable and auditable by all approved members.

The app is built for worldwide use and is deliberately neutral: four launch languages (English, Arabic, Urdu, Hindi), no national flags or country associations in the interface, and currency, decimals and time zone driven by each Room's settings.

2. Problem Statement

Shared-cost groups track money in chat groups, notebooks or memory. This causes:

- Disputes over who paid what and who owes how much
- No proof (receipts, history) when disagreements arise
- Late or inconsistent settlements with no accountability
- Poor access for non-English speakers in multinational households

Generic bill-splitting apps lack Admin-controlled membership, an immutable audit trail, payment confirmation with proof, and true multi-language, multi-currency support. Mess Manager closes this gap.

3. Goals & Success Metrics (KPIs)

Targets are indicative; the Product Owner validates them before development sign-off.

Metric	Target
Settlement acknowledgement rate	⩾ 90% of members acknowledge within 7 days
Expense dispute rate	< 2% of expenses flagged per month
Dispute resolution time	⩾ 80% of flags resolved within 72 hours
Payment confirmation turnaround	⩾ 80% of member payment claims confirmed or rejected by Admin within 48 hours
Push notification delivery	⩾ 95% delivered within 10 seconds
Offline sync success	⩾ 99.5% of queued offline actions sync without manual intervention
Language coverage at launch	100% of UI strings, notifications, emails, PDFs and legal documents in all 4 languages
App stability	⩾ 99% crash-free sessions
Admin approval turnaround	⩾ 80% of join requests actioned within 24 hours

4. Target Users & Personas

Persona	Context	Primary Need
Room Admin (flat, hostel or mess manager)	Manages a shared space, approves members, confirms payments, runs settlements	Full visibility and control with minimal manual reconciliation

Persona	Context	Primary Need
Room Member (roommate, colleague)	Adds and shares expenses, pays dues	Trust that expenses are accurate, visible and fairly split
Family Member	Shares household costs with relatives	Simple tracking without financial ambiguity
Office / Team Lead	Manages pantry, outing or team-fund expenses	Auditable records for reimbursement
Low-connectivity user (worker camps, rural hostels)	Patchy or expensive mobile data	Add and view expenses offline; reliable sync later

5. Scope

5.1 In Scope — Version 1

- Email/password and Google Sign-In, signup consent, optional App Lock (PIN/biometric)
- In-app account deletion with a 2-month audit hold, plus the web deletion page Google Play requires
- Room creation, join by code or invite, Admin approval workflow
- Expense logging with optional receipts, split methods and a permanent audit history
- Pro-rated settlement engine based on active stay period
- Payment claims with mandatory proof, Admin confirmation, and month lock after settlement
- Dispute and flag system
- Need Request system with push notifications
- Offline read and offline expense entry with safe sync
- PDF/CSV reports, including the automatic PDF statement before data deletion
- Four launch languages (English, Arabic, Urdu, Hindi) with full RTL support; four themes
- Ad-supported free model with privacy-compliant consent
- 2-calendar-month data retention, then automatic deletion

5.2 Out of Scope — Version 1

- In-app payment gateway or money transfer (payments are recorded and confirmed, never processed)
- Cross-room currency conversion
- OCR receipt scanning, QR join, AI insights, inventory, voice assistant
- Full VoiceOver / TalkBack support (baseline contrast and font scaling are in v1)
- Organization and multi-family workspaces
- Paid / ad-free subscription tier

6. Recommended Tech Stack

Layer	Choice
Frontend	React Native (Expo), TypeScript strict mode
Backend	Supabase (Edge Functions for settlement, deletion and scheduled jobs)
Database	PostgreSQL with Row Level Security
Authentication	Supabase Auth (email/password + Google OAuth)
Storage	Supabase Storage (private buckets, signed URLs)
Offline store	Local SQLite (e.g. WatermelonDB or expo-sqlite) with an outbox queue
Push Notifications	Expo Push Notifications + in-app notification inbox
Transactional Email	A transactional email service (e.g. Resend or Amazon SES)
PDF Generation	Server-side rendering with embedded Arabic, Urdu (Nastaliq) and Devanagari fonts
Ads	Google AdMob with Google's User Messaging Platform (UMP) for consent
Crash & error reporting	Sentry or Firebase Crashlytics (no personal data in logs)

Layer	Choice
Device security	Expo LocalAuthentication (biometric App Lock), SecureStore for tokens
Version Control	Git + GitHub with protected main branch and CI

7. Authentication & User Management

7.1 Login

- Email + password login, Forgot password (email reset link), Stay signed in (optional), Logout
- Continue with Google (Section 7.3)
- Login attempts are rate-limited; after 5 failed attempts the account is temporarily locked for 15 minutes

7.2 Signup

- Fields: Name, Email, Password, Confirm Password
- Password minimum 8 characters, checked against common-password lists; stored only as a secure hash by Supabase Auth
- Email verification required before joining or creating a Room
- Signup is blocked until the user ticks the mandatory consent box for the Privacy Policy and Terms & Conditions; consent version, date and time are stored
- Date of birth is asked at signup (age gate). Users under 18 need verified parent/guardian consent before the account becomes active (Section 20.4)

7.3 Google Sign-In — Privacy Rules

- Request only the minimum Google permissions: name, email, profile picture. Never contacts, Drive, Gmail, location or any other scope.
- Google never shares the user's Google password with the app; the app stores only the verified email and Google account ID.
- If an email already registered with a password signs in with Google, the accounts link only after the user enters the existing password once. This prevents account takeover.
- A Google-only user can optionally set an app password later from Settings.

- The Privacy Policy states exactly what is received from Google and that it is not used for advertising.

7.4 App Lock (device privacy)

Google Sign-In keeps the user signed in without a password, so anyone holding an unlocked phone could open the app. App Lock solves this:

- Optional PIN, fingerprint or face unlock, set from Settings
- Asked when the app opens and after 5 minutes in the background (configurable)
- Sensitive actions (delete account, change password, Admin removal of members) always require re-authentication, regardless of App Lock

7.5 Member Privacy Inside a Room

- Other members see only display name and profile picture
- A member's email address is visible only to Room Admins
- Receipts are visible only to approved members of that Room

7.6 Language Selection

Available before login; applies to login, signup, legal documents and the whole app. Saved as a user preference. Languages are shown by their own name in their own script, with no flag (Section 16).

7.7 User Profile & Settings

- Update name and profile picture; change password, theme and language
- App Lock settings
- Notification preferences (mandatory events cannot be switched off)
- Ad privacy choices (Section 19)
- **Delete Account** — available directly inside Settings (Section 18.3)

Country selection is not required during signup.

8. Room Management

Every Room has: Room Name, Country, Currency, Time Zone and Settlement Cycle. These settings belong to the Room, are set by an Admin, and apply identically to every member.

8.1 Country, Currency & Time Zone

- When an Admin selects the Room's Country, the currency is auto-selected (ISO 4217) with its standard decimals — e.g. KWD 3, USD 2, JPY 0.
- Time Zone follows the Country. For countries with more than one time zone, the Admin chooses one zone from that country's list (IANA database).
- The selection applies to the whole Room. Every member sees the same Country, Currency and Time Zone on the Room screen.
- Any change is shown to all members: a push notification, an in-app banner, and a permanent audit entry (who changed what, old value, new value, date and time).
- Country appears only as a plain text name from a standard reference list, in alphabetical order. No flags or national symbols (Section 16).

8.2 Currency Change Rule

Currency can be changed only when all three conditions are true:

1. It is the start of a new settlement cycle (for a Monthly cycle: the start of the month, in the Room's time zone)
2. No expense has yet been added in the new cycle
3. The previous cycle is settled and locked (Section 11.5)

Once the first expense of the cycle is added, the currency option is disabled until the next cycle. Each locked cycle keeps its original currency forever in history and reports; amounts are never converted.

Carried-forward dues: if a member still owes money from a locked cycle in the old currency, that balance stays in the old currency and is shown separately until the Admin confirms it paid.

8.3 Time Zone Change Rule

A time-zone change takes effect from the start of the next settlement cycle, so cycle cut-off dates never shift mid-cycle. The new zone is shown to all members as "effective from [date]".

8.4 Room Join Methods

- Room Code
- Email invitation link (includes the app download link)

8.5 Room Approval

Every join request needs Admin approval. Until approved, a user cannot enter the Room, view expenses or participate. A Room Code never bypasses approval.

8.6 Admin Management

Multiple Admins are supported. Admins approve, reject or remove members; promote or demote Admins; configure Room settings and the settlement cycle; confirm payments; resolve disputes; and delete the Room (conditions apply).

- **Last Admin Rule:** the last remaining Admin cannot leave, delete their account, or delete the Room until another Admin is assigned.
- **Room Delete Rule:** a Room cannot be deleted while approved members exist or while any cycle is unsettled.

8.7 Leaving a Room

- Expenses are calculated only for the member's active stay period
- The member's statement for the current cycle is automatically emailed as a PDF
- The member's data follows the 2-month Retention Policy (Section 18)
- A former member must request approval again to rejoin

9. Member Management

Action	Member	Admin
View expenses, receipts, audit history	Yes	Yes
Add expense, upload receipt	Yes	Yes
Edit / delete own expense	Within 30 minutes, open cycle only	Any expense, open cycle only
Claim own payment as paid (with proof)	Yes	Yes
Confirm or reject payment claims	No	Yes
Raise a dispute / flag	Yes	Yes

Action	Member	Admin
Resolve a dispute	No	Yes (not on their own expense — Section 12)
Approve / remove members, manage Admins	No	Yes
Generate settlement, configure cycle and Room settings	No	Yes
Download own reports	Yes	Yes
Download full Room reports	No	Yes

10. Expense Management

Each expense contains: Title, Amount, Category, Expense Date, Paid By, Split Method, Notes (optional) and Receipt (optional).

10.1 Split Methods

- **Equal, pro-rated by stay (default):** shared among members active on the expense date, adjusted for each member's active stay period in the cycle
- **Selected members:** the person adding the expense ticks who shares it (e.g. a personal item bought for two members only)
- **Meal count (decision pending — Section 28):** for food messes; per-meal rate = total food expense ÷ total meals, each member pays meals × rate

10.2 Amounts & Rounding

- All amounts are stored as integers in the currency's minor unit (fils, cents, paise) — never floating point
- Rounding follows the currency's decimals; any leftover minor unit from a split is assigned to the payer, so totals always match exactly

10.3 Receipt Upload

Optional for expenses. Sources: Camera, Gallery. Formats: JPG, PNG, PDF; maximum 10 MB, images compressed on device. Receipts are visible to all approved Room members and stored in a private bucket accessed through short-lived signed links.

10.4 Expense Edit Policy

- Member: may edit or delete their own expense within 30 minutes, measured by the server clock
- Admin: may edit or delete any expense while the cycle is open
- Nobody can edit or delete an expense in a locked cycle (Section 11.5)
- "Delete" is a soft delete: the expense disappears from totals but remains visible in history

10.5 Expense History

Every add, edit and delete is permanently recorded: Action, Old Value, New Value, User, Date and Time (server time, shown in the Room's time zone). History cannot be modified by anyone, including Admins and Uyoon Tech staff.

11. Settlement & Payment Management

A payment counts as Paid only after an Admin confirms it, and every cycle is locked the moment its settlement is generated.

11.1 Settlement Cycle

Configured by Admin only: Daily, Weekly or Monthly (Monthly recommended). Settlement always follows the Room's time zone and currency, and cut-off times are computed on the server.

11.2 Expense Calculation

Each member pays only for their active stay: a member joining on the 10th is charged from the 10th; a member leaving on the 18th is charged until the 18th. Join and leave days count as active days. No member pays before joining or after leaving.

11.3 Settlement Result

Total Room expense, each member's share, amount paid, amount payable or receivable, and payment status. The app also shows a simplified "who pays whom" list that settles all balances with the fewest transfers.

11.4 Payment Claim & Confirmation

1. The paying member taps **Mark as Paid** and enters the amount and method (Bank / Wallet / Cash / Other).
2. Proof is mandatory for every non-cash method: a screenshot or photo of the transfer, or a transaction reference number.
3. For Cash, proof is optional; the Admin's confirmation that cash was received is the record.
4. Status becomes **Awaiting Confirmation**. All Admins are notified.
5. An Admin **confirms** (status becomes Paid or Partially Paid) or **rejects** with a mandatory reason (status returns to Pending).
6. An Admin can also record a payment directly, e.g. cash received in hand; this is logged in the audit trail with the Admin's name.
7. An Admin cannot confirm a payment they claimed themselves; another Admin must confirm it. In a single-Admin Room, the claim is shown to all members as "self-confirmed by Admin".

Payment statuses: Pending · Awaiting Confirmation · Partially Paid · Paid · Rejected. Payment history and proofs are visible to all approved members.

11.5 Settlement Lock

- When the Admin generates a cycle's settlement, that cycle is **locked**: its expenses can no longer be added, edited or deleted by anyone, including Admins.
- Payments against a locked cycle can still be claimed and confirmed until fully paid, because collecting money happens after the lock.
- A genuine mistake found after locking is corrected only by an **adjustment entry** in the current open cycle, linked to the original expense and recorded in the audit trail.
- Locking is enforced in the database, not only in the app, so it cannot be bypassed through offline sync or a modified app.

11.6 Settlement Statement on Leaving

When a member leaves, an automatic PDF email covers: active stay period, expense summary, settlement summary, outstanding amount and receipt summary.

11.7 Settlement Acknowledgement

Every member must acknowledge each settlement notification. Admins see Delivered / Seen / Pending per member.

12. Dispute & Flag System

Any approved member can flag an expense, a payment confirmation or a settlement line they believe is wrong.

- Flag fields: reason (Wrong amount / Not a shared expense / Duplicate / Payment not received / Other), a short note, optional evidence
- All Admins and the person who created the flagged item are notified
- A flagged item shows a visible "Under dispute" badge to all members
- Resolution: an Admin resolves the flag (Accepted → correction made; Rejected → reason given). An Admin cannot resolve a flag on their own expense or payment; another Admin must, or in a single-Admin Room the resolution is marked "resolved by involved Admin"
- An expense under dispute can still be settled; if accepted later, it is corrected by an adjustment entry in the next open cycle
- Disputes and their outcomes are part of the permanent audit history and of the 2-month retention record
- Flag statuses: Open · Under Review · Accepted · Rejected · Withdrawn

13. Need Request System

Any approved member can create a Need Request (e.g. Rice, Oil, Milk, Vegetables, Gas, Cleaning Items). All members get a push notification; any member can accept responsibility, and others see who accepted, which reduces duplicate purchases. When completed, the app offers to turn the purchase into an expense in one tap.

Statuses: Pending · Accepted · Completed · Cancelled.

14. Notification System

Push notifications plus an in-app notification inbox (fallback if push fails) for: Join Request, Join Approval / Rejection, Expense Added / Edited / Deleted, Room settings changed (country, currency, time zone, cycle), Settlement Generated, Payment Claimed,

Payment Confirmed / Rejected, Dispute Raised / Resolved, Need Request Created / Accepted / Completed, Admin Promotion, Member Removed, Member Account Deletion Scheduled.

Settlement notifications include Cycle, Amount, Payment Status and Due Date. Acknowledgement is recorded and Admins see Seen / Not Seen. All notifications follow each recipient's chosen language.

15. Reports & Transparency

- Every member can view and download their own reports; Admins can also download full Room reports
- Formats: PDF and CSV
- Reports include Expenses, Settlements, Payments (with proof references), Disputes and Audit History
- Reports are generated in the requester's language, with correct right-to-left layout for Arabic and Urdu
- Each report carries a generation date and a reference number so its authenticity can be checked

16. Themes, Localization & Neutrality

Mess Manager is a universal app: no language is tied to a country, and no national flag or symbol appears anywhere.

16.1 Neutrality Rules

- Language picker shows each language by its own name in its own script only: English · العربية · اردو · हिन्दी. No flags, no country names next to languages.
- Country lists show plain text names from a standard reference list (ISO 3166), alphabetically, with no flags, maps or emblems.
- Currencies show the ISO code and symbol only (e.g. KWD, INR, USD).
- No maps or borders anywhere in the app, store listing or marketing screenshots, to avoid territorial sensitivities.
- Default illustrations, avatars and category icons are culturally neutral.

16.2 Localization

- The whole app follows the selected language: UI, notifications, emails, PDF reports, Privacy Policy and Terms
- Full right-to-left (RTL) layout for Arabic and Urdu, including mirrored navigation and correct mixing of numbers inside RTL text
- Number display: Western digits by default, with an optional setting for Eastern Arabic digits; stored values never change
- Dates shown in the user's language, calculated in the Room's time zone
- All text is kept in translation files; no text is hard-coded in screens
- Legal documents and app text are translated by professional human translators and reviewed by a native speaker; machine translation is not accepted for release
- Urdu uses a Nastaliq font; Hindi uses a Devanagari font; both embedded in PDFs

16.3 Themes

Four professional themes (including light and dark), changeable any time. All themes meet baseline WCAG contrast and respect the phone's font-size setting.

17. Offline Mode & Sync

Offline mode stays in v1 for poor-network regions; these rules keep it bug-free and keep money calculations correct.

17.1 What Works Offline

Action	Offline
View Room, expenses, balances, history (last synced)	Yes
Add an expense (with receipt photo)	Yes — queued
Edit / delete own expense	Yes — queued, still subject to the 30-minute and lock rules on sync
Create or accept a Need Request	Yes — queued
Mark a payment as paid with proof	Yes — queued

Action	Offline
Approve members, confirm payments, resolve disputes, generate settlement, change Room settings	No — Admin actions need a live connection

17.2 Sync Safety Rules

1. **Outbox queue:** every offline action is saved locally with a unique ID, so retries never create duplicates.
2. **Server decides:** the server re-checks every queued action (permissions, 30-minute window, cycle lock) using server time. The device clock is never trusted.
3. **Locked cycle:** an offline expense whose date falls in a cycle locked meanwhile is not lost; it is moved to the current open cycle as a late entry, and the creator is told.
4. **Conflicts:** if two people changed the same expense, the earlier server-accepted change wins; the other is returned to its author to review, never silently overwritten.
5. **Clear status:** each queued item shows a "Waiting to sync" badge; failures show the reason and a retry button.
6. **Balances offline** are labelled "Estimated — not yet synced"; official totals come only from the server.
7. **Receipts** upload in the background, compressed, resuming after network drops.
8. **Logout warning:** the user is warned before logout or account deletion if unsynced items exist.

17.3 Testing Requirement

Offline flows are tested on low-end Android devices with simulated 2G, network loss mid-sync, airplane mode, wrong device clock and app kill during sync, before every release.

18. Security, Data Retention & Account Deletion

No financial data is kept longer than 2 complete calendar months; before anything is deleted, every affected member receives a PDF record by email.

18.1 Security

- HTTPS (TLS 1.2+) for all traffic; data encrypted at rest

- Row Level Security on every table: a user can read only Rooms they are an approved member of
- Settlement, locking, payment confirmation and deletion run only on the server
- Receipts and proofs in private storage with short-lived signed links
- Audit trail is append-only: no update or delete permission for anyone, including Admins
- Tokens stored in the phone's secure storage; no personal data in crash logs
- An independent security test (Section 25.3) is passed before Play Store release

18.2 Data Retention (2 calendar months)

- Every expense, settlement, payment, dispute and audit record is kept for **two complete calendar months** after the cycle it belongs to is locked.
- **7 days before deletion**, every member of the Room automatically receives an email with a PDF statement of the cycle that will be deleted (expenses, settlements, payments, disputes, audit summary) in their own language, plus an in-app reminder to download reports.
- After the 2 months, the records are deleted from the live database and removed from backups within 30 days. Uyoon Tech does not keep a long-term archive.
- Users are told clearly, in the app and in the Terms, that the emailed PDF is their permanent copy.

18.3 Account Deletion — In-App (primary)

Settings → Account → Delete Account. The full flow:

1. The app shows what happens: login ends immediately; financial records stay for 2 months only for accounts and dispute purposes; a final PDF will be emailed; the action cannot be undone.
2. Warnings appear if the user has unsynced offline items, unpaid dues or open disputes. These do not block deletion.
3. The **last Admin** of a Room must first assign another Admin (the only blocking rule).
4. The user re-enters their password (or re-confirms with Google) and confirms.
5. **Immediately:** login is disabled; profile picture, App Lock data, device tokens and preferences are deleted; the user leaves all Rooms; their active stay ends today; all Room members are notified.

6. **During the 2-month hold:** the user's name appears in Room history as "[Name] (account deleted)". Records are read-only and used only for accounts, audit and dispute resolution, never for ads or marketing.
7. **At the end of the hold:** the deleted user (at their last email) and all members of their former Rooms receive the final PDF statement by email; then all remaining data linked to that person is permanently deleted, including the email address.

18.4 Account Deletion — Web Page (required by Google Play)

Google Play requires both an in-app deletion path and a web link where users can request deletion, so someone who has already uninstalled the app does not need to reinstall it. The in-app option stays the main path. The web page is minimal:

- A page on the Uyoon Tech website: the user enters their email, receives a confirmation link, and confirms
- The same lifecycle as 18.3 then runs
- The page link goes into the Play Console Data safety form and the Privacy Policy

18.5 User Data Rights

Users can, from Settings: download all their data (PDF/CSV), correct their profile, withdraw ad-personalisation consent, and delete their account. Requests by email to the Grievance Officer (Section 20) are answered within 30 days.

19. Ads & Monetization

Mess Manager is free for all users and earns revenue from ads. The founder's specific ad approach is to be added (Section 28); whatever it is, the rules below are mandatory.

19.1 Placement Rules

- No ads on screens where money or trust decisions happen: payment claim, proof upload, payment confirmation, settlement generation, dispute, account deletion, login and signup
- No ad ever covers or sits next to an action button (prevents accidental clicks, an ad-network policy violation)
- No ads in emails, PDF reports or push notifications
- Full-screen ads, if used, are rare (frequency-capped) and never interrupt a task in progress

19.2 Content Rules

Block sensitive ad categories in the ad network settings: gambling, alcohol, dating, adult, political, religious and get-rich-quick ads. This protects the app's neutral, family-safe position across all markets.

19.3 Privacy & Consent

- Financial data, receipts, Room names and expense details are **never** shared with advertisers or used for ad targeting
- Users in the EEA, UK and Switzerland see a certified consent screen before any personalised ad; elsewhere, users can switch to non-personalised ads anytime in Settings
- Ads are limited to non-personalised for any user who has not consented, and always for users under 18 (Section 20.4)
- Ad SDK data collection is disclosed in the Privacy Policy and the Play Store Data safety form

19.4 Future

An optional paid ad-free tier can be added later without changing the data model.

20. Legal & Compliance

The Privacy Policy and Terms & Conditions are drafted to a professional standard, then reviewed and signed off by a qualified Indian technology lawyer before launch. The goal is to protect Uyoon Tech Solutions and its founder by stating clearly what the app does and does not do.

20.1 Laws to Address (final list confirmed by counsel)

- **India:** Digital Personal Data Protection Act 2023 and its Rules; Information Technology Act 2000 and related rules on reasonable security practices and grievance redressal
- **European Union / UK:** GDPR and UK GDPR (if the app is offered there)
- **Gulf region:** personal data protection laws of Saudi Arabia, UAE, Bahrain, Qatar and Kuwait
- **Other key markets:** e.g. US state privacy laws, Pakistan, Bangladesh — as launch markets are confirmed

- **Platform policies:** Google Play Developer Program Policies, User Data policy, AdMob policies

20.2 Protective Terms & Conditions (key clauses)

- Mess Manager is a **record-keeping tool only**. It does not hold, transfer or process money and is not a bank, wallet or payment service.
- Users and Room Admins are solely responsible for the accuracy of what they enter and for settling money among themselves.
- Uyoon Tech is **not a party** to any dispute between members; the in-app dispute feature is a convenience only.
- Records are kept for 2 calendar months only; the emailed PDF is the user's own copy, and Uyoon Tech has no obligation to keep data longer.
- Service is provided "as is"; limitation of liability to the maximum extent permitted by law; no liability for losses from user error, network failure or third-party services.
- Users under 18 may use the app only with verified parent/guardian consent; the parent/guardian accepts these Terms on the child's behalf and is responsible for the child's use.
- Acceptable use: no illegal transactions, money laundering, gambling pools or abuse.
- Uyoon Tech may suspend accounts that misuse the service.
- Governing law: India; jurisdiction: courts of [city — to be decided], with users' mandatory local consumer rights preserved.
- Terms can change with notice inside the app; continued use means acceptance; material changes need fresh consent.

20.3 Privacy Policy (key contents)

What data is collected (account, Room, expenses, receipts, device and ad identifiers), why, legal basis, who it is shared with (Supabase hosting, email service, ad network, crash reporting), where it is stored, the 2-month retention and deletion process, user rights, ad choices, security measures, children's policy, and a named **Grievance Officer** with email and response time.

20.4 Children & Age Policy

The app is open to all ages, including children of about 10–12 years (for example, in families or hostels). Indian law treats everyone under 18 as a child and requires verifiable parent consent and no targeted ads for them; similar child-protection rules apply abroad. The

following protections are mandatory:

- **Age gate:** date of birth at signup, asked neutrally (no hint of a "right" answer)
- **Parental consent:** a user under 18 enters a parent/guardian's email; the parent verifies and consents before the account is activated. The parent can view the child's Rooms, revoke consent and delete the child's account
- **No tracking or targeted ads:** under-18 users see only non-personalised, child-safe ads from child-safe certified ad networks, and no analytics profiling
- **Minimal data:** no profile picture required; a child's email is never shown to anyone in the Room, including Admins (only display name)
- **Room safety:** a minor cannot create a Room or become an Admin (default — see Section 28); a minor joins only with Admin approval as usual
- **No free-text contact:** the app has no chat or direct messaging, so strangers cannot message a child; notes and dispute text are visible only inside the Room
- **Google Play:** declare the target audience to include children; this brings the app under Google Play's Families policy, so only Families-compliant ad SDKs and features are used
- **Legal documents:** a child-friendly summary of the Privacy Policy in simple words, in all 4 languages

20.5 Play Store Publishing Requirements

- Publish under an **Organization** developer account for Unifair Trading Pvt. Ltd.; apply for the company's D-U-N-S number now, as it can take up to 30 days. Organization accounts are exempt from the 12-tester, 14-day closed-test rule that applies to new personal accounts.
- Complete the Data safety form, including ads, data deletion and the deletion web link
- Privacy Policy hosted at a public web address
- Target the Android API level Google requires at the time of submission

21. Non-Functional Requirements

Category	Requirement
Performance	Expense list and settlement summary load within 2 seconds on 4G; usable on 2 GB RAM Android phones

Category	Requirement
Availability	Backend targets 99.5% monthly uptime
Offline	Rules in Section 17; no data loss on network failure or app kill
Scalability	A Room supports at least 50 active members without redesign
Accuracy	All money as integer minor units; settlement totals always equal the sum of shares exactly
Localization	All text externalised; full RTL for Arabic and Urdu
Security	Section 18.1; independent security test passed before release
Privacy	Minimum data collected; no financial data to advertisers
Accessibility	WCAG-aligned contrast and font scaling in v1; full screen-reader support post-v1
App size	Install size under 40 MB
Compliance	Section 20

22. Business Rules

- Admin approval is mandatory for every Room join; a Room Code never bypasses it; former members need fresh approval.
- The last Admin cannot leave, delete their account or delete the Room until another Admin exists.
- A Room cannot be deleted while approved members exist or a cycle is unsettled.
- Country determines currency; Room settings apply to all members and every change is shown to all.
- Currency can change only at the start of a cycle, before its first expense, with the previous cycle locked.
- A time-zone change takes effect from the next cycle.
- A member pays only for their active stay period.

- Receipt upload for expenses is optional; payment proof is mandatory for non-cash payments.
- A payment is Paid only after Admin confirmation; an Admin cannot confirm their own payment where another Admin exists.
- A cycle is locked when its settlement is generated; corrections only by adjustment entries.
- Expense and audit history are immutable.
- Any member can flag; an Admin cannot resolve a flag on their own item where another Admin exists.
- Push notifications are mandatory for defined events.
- Reports: members download their own; Admins download the full Room.
- Data is retained for 2 complete calendar months, a PDF is emailed to affected members, then data is permanently deleted.
- Account deletion is available in Settings and via the web page; financial records are held 2 months for audit only.
- Privacy Policy and Terms acceptance are required before signup; users under 18 also need verified parent/guardian consent.
- No national flags or symbols anywhere in the app.
- No ads on money, trust or account screens.

23. Requirement Prioritization (MoSCoW)

v1.2 moves languages, notifications, reports, offline mode, payment confirmation, month lock, disputes and account deletion to Must, because the launch scope and success criteria depend on them.

Feature Area	Priority
Authentication, Google Sign-In, signup consent, age gate + parental consent	Must
Account deletion (in-app + web page) and 2-month retention	Must
Room creation, join, Admin approval, Room settings rules	Must

Feature Area	Priority
Expense add / edit / delete, split (equal + selected members), history	Must
Settlement engine (pro-rated) and month lock	Must
Payment claim with proof and Admin confirmation	Must
Dispute & flag system	Must
Security: RLS, HTTPS, audit trail	Must
Push notifications + in-app inbox	Must
4 languages with RTL, neutral UI	Must
Offline read + offline entry with safe sync	Must
Reports (PDF/CSV) and pre-deletion PDF emails	Must
Ads with consent	Must
Need Request system	Should
Settlement acknowledgement (Seen / Pending)	Should
App Lock (PIN / biometric)	Should
Four themes	Should
Meal-count split	Could (pending decision)
"Who pays whom" simplified transfers	Could
OCR, QR join, AI insights, inventory, voice, full screen reader, workspaces, cross-room currency, ad-free tier	Won't (v1)

24. Risks & Assumptions

Risk	Impact	Mitigation
Offline sync creates duplicate or wrong entries	High	Outbox with unique IDs, server-side validation, conflict review (Section 17)

Risk	Impact	Mitigation
Users lose records after 2 months and blame the app	Medium	7-day advance PDF email, in-app reminders, clear Terms
Payment proofs are fake screenshots	Medium	Admin confirmation required; proofs and confirmations auditable; Terms place responsibility on users
Single-Admin Rooms self-confirm payments	Medium	Visible "self-confirmed" label; encourage a second Admin
Ads breach policy or privacy law	High	Placement and content rules, consent screen, no financial data to ad network (Section 19)
Poor translations damage trust	Medium	Professional translators + native-speaker review
Time-zone / DST errors shift cut-offs	Medium	IANA database, server-side computation, changes effective next cycle
Push delivery failures	Low	In-app notification inbox fallback
Clock manipulation to extend 30-minute edit window	Low	Server timestamps only
D-U-N-S / Play verification delays launch	Medium	Apply for D-U-N-S immediately, in parallel with development

Assumptions: users of all ages (minors with parent consent) have basic smartphone literacy; internet is available at least intermittently; the Room Admin is the trusted day-to-day contact; supported countries, currencies and time zones come from maintained standard lists.

25. Delivery Plan & Future Roadmap

25.1 Indicative Delivery Timeline (1–2 experienced developers)

Phase	Work	Duration
0	PRD sign-off, screen designs, D-U-N-S and Play Console setup	1–2 weeks
1	Auth, Rooms, expenses, settlement, lock, payments, audit	5–6 weeks
2	Offline sync, disputes, deletion lifecycle, notifications	2–3 weeks
3	4 languages + RTL, reports, ads, themes, Need Requests	2–3 weeks
4	Security test, QA on low-end devices in all 4 languages	2 weeks
5	Play Store review and staged rollout	1–2 weeks

Total: about 13–18 weeks. Offline mode keeps it slightly longer than the v1.1 estimate.

25.2 Parking Lot (post-v1)

OCR receipt scanner · QR Room join · AI expense insights · inventory management · voice assistant · full VoiceOver / TalkBack · advanced analytics · organization and family workspaces · cross-room currency conversion · paid ad-free tier · iOS release.

25.3 Security Check Before Release

An independent tester checks: RLS (no cross-Room data access), authentication and account linking, locked-cycle bypass attempts, offline sync tampering, storage link exposure, and the OWASP Mobile Top 10.

26. Technical Guidelines (Developer Reference)

Database

- PostgreSQL (Supabase), UUID primary keys, foreign keys, Row Level Security on every table
- Money as integer minor units with an ISO 4217 currency code on every amount
- Automatic created/updated timestamps from the server
- Append-only audit table (no UPDATE/DELETE grants)
- Cycle lock enforced by database rules, not only app logic
- Scheduled server jobs for: cycle cut-off, 7-day pre-deletion emails, retention deletion, account-deletion holds
- Idempotency keys on every write so offline retries are safe

Backend APIs

Secure, authenticated APIs for: Authentication, Account Deletion, Room Management, Member Management, Expenses, Settlement, Payments & Confirmation, Disputes, Need Requests, Notifications, Reports, File Upload, Audit Logs.

Development Standards

Clean, modular architecture · TypeScript strict mode · reusable components · responsive UI · externalised translations with RTL · dark/light themes · secure coding · Git with code review and CI · automated tests for settlement maths, locking, permissions and sync · well-documented code.

27. Success Criteria

- All approved PRD requirements implemented; no feature removed without Product Owner approval
- Settlement totals match exactly in every currency precision (0, 2, 3 decimals)
- No locked cycle can be changed through the app, offline sync or the API
- App works correctly in all 4 languages, including RTL
- Offline tests in Section 17.3 pass
- Account deletion and retention emails work end to end
- Independent security test passed

- Privacy Policy and Terms reviewed by a lawyer and published
- App approved and live on Google Play; source code documented; deployment instructions provided

28. Open Decisions for the Founder

- [ ] Describe the planned ad approach ("khaas tareeka") so Section 19 can be finalised
- [ ] Include the meal-count split in v1, or keep it for a later release?
- [ ] City for legal jurisdiction in the Terms
- [ ] Name and email of the Grievance Officer
- [ ] Company website domain for the Privacy Policy and the account-deletion web page
- [ ] Confirm minors cannot create Rooms or become Admins, and whether there is a minimum age (e.g. 10)
- [ ] Launch markets for the first release (worldwide, or a first set of countries)

Product Manifesto

Mess Manager is built on one principle: every shared expense should remain transparent, verifiable and fair for every member.